

Equivalence Partition TCI Table

TCI	nights	guestAge	isArResident	hasVetDiscount	Outcome
EP1	Integer.MIN_VALUE ≤ nights < 1	13 ≤ guestAge ≤ 64	false	false	NightReservationException
EP2	1 ≤ nights < 15	13 ≤ guestAge ≤ 64	false	false	50 * nights
EP3	15 ≤ nights ≤ Integer.MAX_VALUE	13 ≤ guestAge ≤ 64	false	false	NightReservationException
EP4	1 ≤ nights < 15	Integer.MIN_VALUE ≤ guestAge < 0	false	false	GuestAgeReservationException
EP5	1 ≤ nights < 15	0 ≤ guestAge < 13	false	false	50 * nights * 0.50
EP6	1 ≤ nights < 15	64 < guestAge ≤ Integer.MAX_VALUE	false	false	50 * nights * 0.80
EP7	1 ≤ nights < 15	13 ≤ guestAge < 65	true	false	(50 * nights) - 10
EP8	1 ≤ nights < 15	13 ≤ guestAge < 65	false	true	(50 * nights) * 0.9
EP9	1 ≤ nights < 15	13 ≤ guestAge < 65	true	true	((50 * nights) - 10) * 0.9

Boundary Value Analysis TCI Table

TCI	nights	guestAge	isArResident	hasVetDiscount	Expected Outcome
BV1	min_nights - 1	nominal_adult	false	false	NightReservationException
BV2	min_nights	nominal_adult	false	false	50 * nights
BV3	min_nights + 1	nominal_adult	false	false	50 * nights
BV4	nominal_valid	nominal_adult	false	false	50 * nights
BV5	max_nights - 1	nominal_adult	false	false	50 * nights
BV6	max_nights	nominal_adult	false	false	50 * nights
BV7	max_nights + 1	nominal_adult	false	false	NightReservationException
BV8	nominal_valid	min_age - 1	false	false	GuestAgeReservationException
BV9	nominal_valid	min_age	false	false	50 * nights * 0.50
BV10	nominal_valid	min_age + 1	false	false	50 * nights * 0.50
BV11	nominal_valid	child_upper_bound	false	false	50 * nights * 0.50
BV12	nominal_valid	adult_lower_bound	false	false	50 * nights
BV13	nominal_valid	adult_upper_bound	false	false	50 * nights
BV14	nominal_valid	senior_lower_bound	false	false	50 * nights * 0.80
BV15	nominal_valid	senior_lower_bound + 1	false	false	50 * nights * 0.80

Decision TCI Table

Causes	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15
nights < 1	T	F	F	F	F	F	F	F	F	F	F	F	F	F	F
nights > 14	F	T	F	F	F	F	F	F	F	F	F	F	F	F	F
guestAge < 0	—	—	T	F	F	F	F	F	F	F	F	F	F	F	F
0 ≤ guestAge ≤ 12 (Child)	—	—	F	T	T	T	T	F	F	F	F	F	F	F	F
13 ≤ guestAge ≤ 64 (Adult)	—	—	F	F	F	F	F	T	T	T	T	F	F	F	F
guestAge ≥ 65 (Senior)	—	—	F	F	F	F	F	F	F	F	F	T	T	T	T
isArResident == true	—	—	—	F	T	F	T	F	T	F	T	F	T	F	T

hasVetDiscount == true	—	—	—	F	F	T	T	F	F	T	T	F	F	T	T
Effects															
Throw NightReservationException	X	X													
Throw GuestAgeReservationException			X												
Calculate Base (\$50 * nights)				X	X	X	X	X	X	X	X	X	X	X	X
Apply Child Discount (50% off)				X	X	X	X								
Apply Senior Discount (20% off)												X	X	X	X
Apply Resident Discount (-\$10)					X		X		X		X		X		X
Apply Vet Discount (10% off)						X	X			X	X			X	X